Lecture 10

Some Specieal Discrete Probability Distributions

27.04.2026

Outline

Bernoulli distribution

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- Found by Swiss mathematician Jacob Bernoulli.

Examples of Bernoulli distributions

- Toss a fair coin and obtain a head. $p = 0.5$.
- Roll a fair 6-sided die and obtain a 6. $p = 1/6$.
- Earn an A for this class. $p \in [0, 1]$.

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Find the mean and variance of $X \sim \text{Ber}(p)$.

$$E(X) = 1 \times p + 0 \times (1 - p) = p$$

$$Var(X) = E(X^2) - (E[X])^2 = 1^2 \times p + 0^2 \times (1 - p) - p^2 = p - p^2 = p(1 - p)$$

Recap

Variance σ^2

- **For all random variable**

$$Var(X) = E[(X-\mu)^2] = E[X^2] - \mu^2$$

- **Linear function**

$$Var(aX+b) = a^2 Var(X)$$

- **Constants**

$$Var(c) = 0$$

Standard deviation

$$SD(X) = \sqrt{Var(X)}$$

Bernoulli distribution

$$p(1) = p, \quad p(0) = 1-p$$

$$\mu = p, \quad \sigma^2 = p(1-p)$$

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- The number of 6 obtained when roll four fair 6-sided dice simultaneously. $n = 4, p = 1/6.$
- The number of A's students will earn in this semester. $n =$ number of classes you're taking, p varies by classes...
  $\Longrightarrow$ not really a Binomial random variable!

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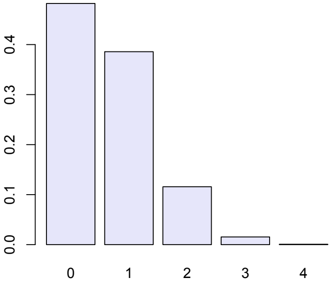

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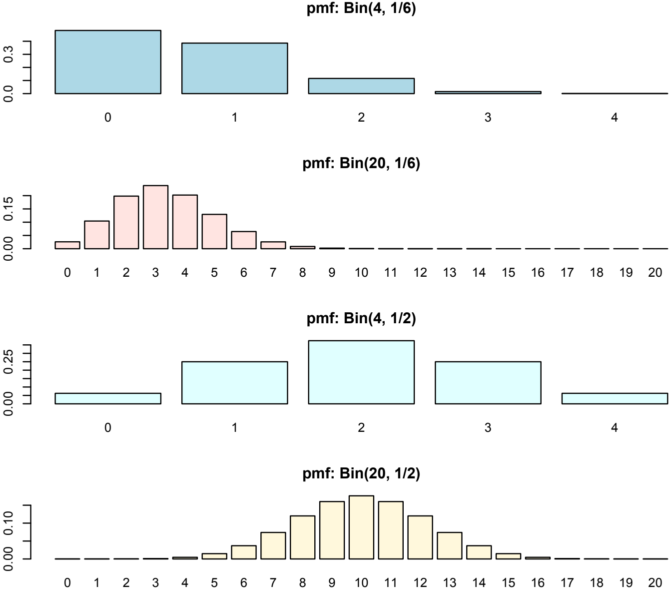
1. the trials are independent
2. the number of trials, n, is fixed
3. each trial outcome can be classified as a *success* or a *failure*
4. the probability of success, p, is the same for each trial (For this example we need to assume that.)

Example

I drink a cup of coffee everyday. About 80% of the time I buy coffee from coffee shop A, about 20% of the time from the coffee shop B. Compute the probability I drink 4 cups of coffee shop B coffee in 10 days.

Pmf of Binomial distribution: uni-modal



Binomial pmf is valid

- Positive: for any $x \in \mathbb{R}$
$$p(x) \geq 0$$
- Total one:
$$\sum_{k=0}^{n} p(k) = 1$$

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Recall the Binomial Theorem $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}$

$$\sum_{k=0}^{n} p(k) = \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} = [p + (1-p)]^n = 1^n = 1$$

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- If we have independent Bernoulli random variable's $X_1, X_2, \ldots, X_n$ with the same probability of success p, then their sum has a Binomial distribution

$$X = X_1 + X_2 + \cdots + X_n \sim \mathsf{Bin}(n, p)$$

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$$E[X] = np = 4 \times (1/6) = 2/3$$
$$Var(X) = np(1 - p) = 4 \times (1/6) \times (5/6) = 5/9$$

Recap

Binomial distribution $X \sim \text{Bin}(n, p)$

$$p(k) = P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, k = 0, 1, \ldots, n$$

- $E(X) = np$
- $Var(X) = np(1-p)$

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Geometric distribution

A gambler plays at a roulette table and alway bet on red until he wins...
In each round, his chance of winning is $18/38 = 0.47$.
Let X denote the number of rounds he plays.

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- Note the difference between Geometric distribution and Binomial distribution! Eg: whether the total number of trials is fixed.

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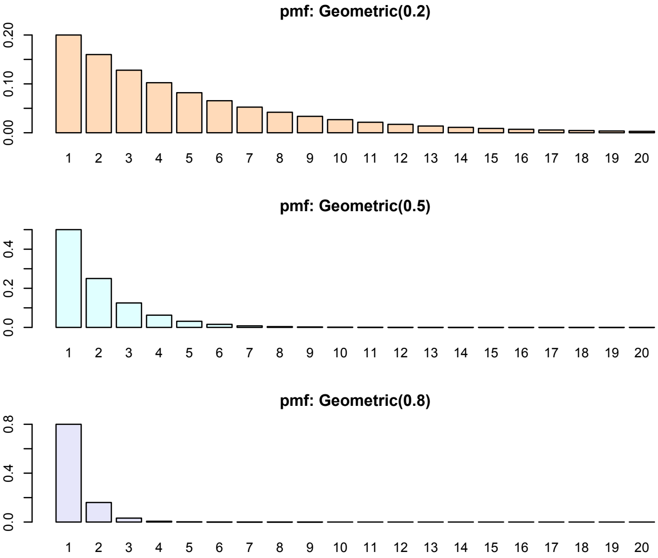
$$\begin{aligned} P(X > n + k \mid X > n) &= \frac{P(X > n + k)}{P(X > n)} \\ &= \frac{(1-p)^{n+k}}{(1-p)^n} = (1-p)^k = P(X > k) \end{aligned}$$

Pmf of Geometric distribution



Negative Binomial distribution

Definition

Denote random variable X that takes value in $\{1, 2, \ldots\}$ as having a Negative Binomial distribution with parameter $p \in (0, 1)$ if its pmf is

$$X \sim NB(r, p) \iff p(k) = \binom{k-1}{r-1}(1-p)^{k-r}p^r, \quad k = r, r+1, \ldots$$

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- Well-defined (validness of pmf).
- Connection between Negative Binomial and Geometric distributions

$$X \sim \text{Geometric}(p) \iff X \sim \text{NB}(1, p)$$

Note: Proof of $\sum_{k=r}^{\infty} p(k) = 1$ for Negative Binomial distribution is not required.

Properties of Neg Binom $p(k) = \binom{k-1}{r-1}(1-p)^{k-r}p^r$

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- Variance

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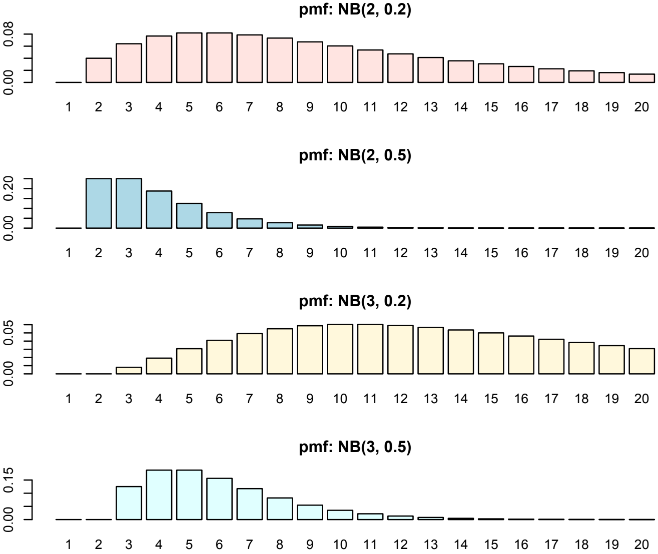
Properties of Neg Binom $p(k) = \binom{k-1}{r-1}(1-p)^{k-r}p^r$

- Mean
$$E[X] = \frac{r}{p}$$
- Variance
$$Var[X] = \frac{r(1-p)}{p^2}$$
- Recall that for Geometric(p),
$$E(X) = \frac{1}{p}, Var(X) = \frac{1-p}{p^2}$$

Pmf of Negative Binomial distribution



Outline

Poisson distribuiton

The Poisson distribution is a discrete probability distribution that models the number of times a random event occurs within a fixed interval of time, space, or volume.

- Number of yeast cells in a fluid sample can be seen under a microscopes.
- The Poisson distribution was used in 1898 to count the number of soldiers in the Prussian Army who died accidentally from horse kicks
- The number of emergency room arrivals during a night shift.
- The number of requests a server receives per second.

Poisson distribution

Definition

Denote random variable X that takes value in $\{0, 1, 2, \dots\}$ as having a Poisson distribution with parameter λ if its pmf is

$$X \sim P(\lambda) \iff p(k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

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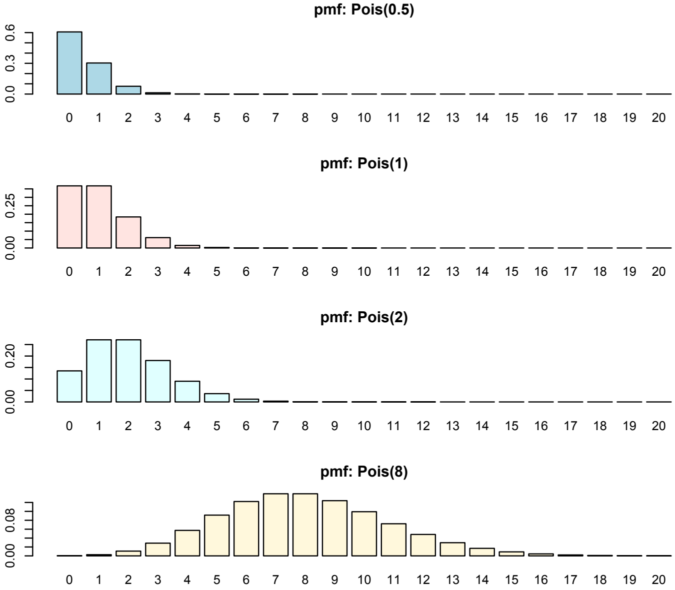
- Usually is used to model "the number of xxx occur". Hence lower bound is 0, no upper bound.

Pmf of Poisson distribution



Properties of Poisson distribution $p(k) = e^{-\lambda} \ \frac{\lambda^k}{k!}$

- Well-defined (validness of pmf): non-negative, and

$$\sum_{k=0}^{\infty} p(k) = 1$$

- Taylor Series

$$f(x) = f(x_0) + \frac{x - x_0}{1!} f'(x_0) + \frac{(x - x_0)^2}{2!} f''(x_0) + \cdots$$

- Use Taylor series to verify that the Poisson distribution is well-defined

$$e^{\lambda} = 1 + \lambda + \frac{\lambda^2}{2!} + \cdots + \frac{\lambda^k}{k!} + \cdots$$

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- Mean

$$E[X] = \lambda$$

- Variance

$$Var[X] = \lambda$$

Suppose ÇÜ students have 1.73 siblings on average. What's the probability that a randomly selected student has at most one sibling?

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$$X \sim \mathsf{P}(\lambda), E[X] = 1.73 \Longrightarrow \lambda = 1.73$$

$$\begin{aligned} P(X \leq 1) &= p(0) + p(1) \\ &= e^{-\lambda}\frac{\lambda^0}{0!} + e^{-\lambda}\frac{\lambda^1}{1!} \\ &= 0.1773 + 0.3067 = 0.4840 \end{aligned}$$

Exercise

A website receives an average of 200 hits per **hour**. Assuming that hits occur randomly and independently at any time, find the probability of getting at least 1 hit in one **minute**.

Exercise

Random variable X follows a Poisson distribution. Given that $p(0) = P(X = 0) = 1/4$, find $Var(2X - 3)$.

Review: discrete distributions

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NegBin(r,p)	$\{r,r+1,\ldots\}$	$\binom{x-1}{r-1}(1-p)^{x-r}p^r$	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$